AVENIRE

Notes — Product Specification

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Avenire Notes is a markdown-native, AI-augmented note-taking module within the Avenire learning platform. It is designed for the student who wants to capture thought at speed, build a connected knowledge graph, and get tutoring assistance inline — without the cognitive overhead of a database-first tool like Notion.

1. Vision & Goals

Avenire Notes is built on a single principle: the path from thought to written note should be frictionless. It takes the best of Obsidian's speed and linked-thought model, Notion's block flexibility, and adds a layer that neither has — a deeply integrated AI tutor that lives inside the document.

Design Principles

- **Speed over structure** — A new note should be one keystroke away. No templates, no setup.
- **Markdown-native, WYSIWYG rendering** — Write in markdown; see formatted output. Syntax visible only at cursor (Obsidian live preview model).
- **Linked thought** — [[wiki-links]] are first-class. Every note exists in a graph, not a folder.
- **AI as collaborator** — Deep Tutor lives inside the editor — explain, quiz, continue, or rewrite, directly in context.
- **Real-time collaboration** — Multiple users and AI can write in the same document simultaneously via CRDT sync.

2. Scope

This spec covers Phase 1 — the core editing experience. Collaboration backend, graph view, and mobile are Phase 2.

In Scope (Phase 1)

- Tiptap-based rich text editor with full markdown support
- Cursor-aware live preview (ProseMirror decoration layer)
- Bubble menu for inline formatting
- Slash command menu for block insertion
- [[Wiki-link]] picker for note linking
- Block types: headings, paragraphs, lists (bullet/numbered/todo), code, blockquote, table, math (KaTeX), image, file, callout, divider, TOC, horizontal rule
- Inline marks: bold, italic, underline, strikethrough, highlight, code, color, link
- AI panel: Deep Tutor explain, continue, quiz, rewrite
- Note sidebar with search and quick-create
- Word count, cursor position statusbar

Out of Scope (Phase 2+)

- Real-time multiplayer (Hocuspocus / Y.js backend)
- Graph view of linked notes
- Mobile / offline support
- Version history and tracked changes
- PDF / DOCX export
- Nested pages (Notion-style)

3. Technical Stack

3.1 Frontend Editor

Package	Purpose
Editor core	Tiptap v2 (React) — ProseMirror-based, headless, fully extensible
Markdown I/O	@tiptap/extension-markdown — serialize/deserialize .md files
Collab (Phase 2)	@tiptap/extension-collaboration + Y.js CRDT

Collab cursors	@tiptap/extension-collaboration-cursor — per-user caret colour
Tables	@tiptap/extension-table + row/header/cell extensions
Math	@tiptap/extension-mathematics — KaTeX inline + block
Task lists	@tiptap/extension-task-list + task-item
Bubble menu	@tiptap/extension-bubble-menu — floating selection toolbar
Slash commands	@tiptap/suggestion — triggers on '/' keypress
Wiki links	@tiptap/suggestion — second instance, triggers on '[['
Color / highlight	@tiptap/extension-color + text-style + highlight
Image	@tiptap/extension-image — with resize handle
Framework	Next.js (App Router) + React 18
Styling	Tailwind CSS + CSS variables for theming

3.2 Backend

Layer	Solution
Collab server	Hocuspocus (@hocuspocus/server) — WebSocket, Y.js aware
Document store	PostgreSQL via @hocuspocus/extension-database
AI	Vercel AI SDK
File storage	S3-compatible (images, file attachments)
Runtime	Node.js / Bun on existing Avenire backend

4. Core Feature: Cursor-Aware Live Preview

The central UX differentiator. The editor renders markdown as formatted output at all times — but when the cursor enters a marked range (e.g., **bold**), the raw syntax characters appear dimmed in monospace, exactly like Obsidian's Live Preview mode.

How it works

Implemented as a ProseMirror Plugin using the Decorations API. On every editor state transaction, the plugin walks the document tree, finds all inline mark ranges matching known markdown patterns, and applies two sets of decorations:

- **Always-on** — Apply formatting styles to the content range (bold, italic colour, etc.) so it renders correctly at all times.
- **Cursor-conditional** — Apply a dim monospace class to the syntax characters (**, *, ==, ~~, `) only when selection.from falls within that mark's range.

This same decoration approach applies to heading markers (#, ##, ###), blockquote markers (>), and list markers (-, 1.) — they appear as dim syntax when the cursor is on that line and are hidden otherwise.

Patterns covered

Syntax	Behaviour
****text****	Bold — renders as bold, ** visible at cursor
text	Italic — renders as italic, * visible at cursor
~~text~~	Strikethrough, ~~ visible at cursor
==text==	Highlight (yellow bg), == visible at cursor
`text`	Inline code (mono, tinted), backtick visible at cursor
[[Note Title]]	Wiki link (blue underline), [[ ]] visible at cursor
# / ## / ###	Heading markers — visible only on cursor line
> text	Blockquote marker — visible only on cursor line
- / 1.	List markers — visible only on cursor line

5. Block Types

5.1 Text Blocks

Block	Description	Trigger
Paragraph	Default block type	Auto on Enter
Heading 1	Page-level title	/h1 or # + space
Heading 2	Section heading	/h2 or ## + space
Heading 3	Subsection heading	/h3 or ### + space
Blockquote	Indented, styled quote	/quote or > + space

Code Block	Monospace, syntax highlight	/code or ``` + Enter
Divider	Horizontal rule	/divider or --- + Enter

5.2 List Blocks

Block	Description	Trigger
Bullet List	Unordered list, nestable with Tab	/ul or - + space
Numbered List	Ordered list, nestable with Tab	/ol or 1. + space
To-do List	Checkboxes, toggleable	/todo or [] + space

5.3 Rich Blocks

Block	Description	Trigger
Table	Editable grid with resize	/table
Math (Inline)	KaTeX expression inline	/math-inline or $...$
Math (Block)	KaTeX display equation	/math-block or $$ + Enter
Image	Upload or URL, with caption	/image
File Attachment	Any file type, shows icon + name	/file
Table of Contents	Auto-generates from headings	/toc
Callout (Info)	Blue info box	/callout-info
Callout (Warning)	Yellow warning box	/callout-warn
Callout (Danger)	Red danger box	/callout-danger

6. Inline Marks

Mark	How to apply
Bold	**text** · Bubble menu B · Cmd+B
Italic	*text* · Bubble menu I · Cmd+I
Underline	Bubble menu U · Cmd+U
Strikethrough	~~text~~ · Bubble menu ~~S~~

Highlight	==text== · Bubble menu ▣ · 7 colour swatches
Inline Code	`text` · Bubble menu </>
Text Color	Bubble menu colour dot → picker
Link	[text](url) · Bubble menu ⌘ · prompts for URL
Wiki Link	[[Note Title]] → triggers picker popover

7. Menus

7.1 Bubble Menu

Appears on text selection. Floats centred above the selection range. Hides on cursor collapse or click-away.

Contents (left to right): B · I · U · ~~S~~ · Highlight · Separator · Code · Link · Separator · Color · Separator · H1 · H2 · H3

7.2 Slash Command Menu

Triggered by typing '/' at the start of a block or on an empty line. A floating panel opens below the cursor, showing categorised block options. Keyboard-navigable (↑↓ Enter Esc). Typing filters options by name.

Categories: Text · Lists · Blocks · Media · Callouts · AI

7.3 Wiki Link Picker

Triggered by typing '[[']. A compact popover appears below the cursor showing notes whose titles match the typed query. Selecting an item inserts a wiki link node and closes the picker. Esc cancels.

8. Deep Tutor Integration

Deep Tutor is Avenire's AI tutor. Inside Notes, it is available in two ways:

8.1 Slash Commands

- **/explain** — Sends the current note's text to Claude with an explanation prompt. Response is inserted as an Info callout below the cursor.
- **/continue** — Continues the document from the cursor position, matching tone and style.
- **/quiz** — Generates 3-5 questions based on the note content, inserted as a To-do block.
- **/rewrite** — Rewrites the selected block for clarity, replacing in-place.

8.2 AI Panel (Bottom Bar)

A persistent panel at the bottom of the editor, toggled via the top bar. Contains a text input for freeform prompts with full note context. Response streams into the panel and can be inserted into the document with one click.

> *All AI calls send a maximum of 1,200 characters of note context as a system prompt prefix. Deep Tutor is instructed to respond in plain text formatted for inline insertion.*

9. Tiptap Extension Architecture

All editor capabilities are composed as discrete Tiptap extensions. The following is the full extension list and their responsibilities.

Extension	Responsibility	Source
StarterKit	P, H1-3, bold, italic, strike, code, blockquote, hardbreak, history	Core
Markdown	Serialize/deserialize .md; enable markdown shortcuts	@tiptap/extension-markdown
Table + Row + Header + Cell	Full table editing with keyboard nav	@tiptap/extension-table
TaskList + TaskItem	Checkbox lists	@tiptap/extension-task-list
Mathematics	KaTeX inline + block	@tiptap/extension-mathematics
Image	Image node with caption	@tiptap/extension-image
Color + TextStyle	Font color mark	@tiptap/extension-color
Highlight	Background highlight mark	@tiptap/extension-highlight
BubbleMenu	Floating selection toolbar	@tiptap/extension-bubble-menu
FloatingMenu	Slash menu anchor	@tiptap/extension-floating-menu
Suggestion (slash)	/ command menu logic	@tiptap/suggestion

Suggestion (wiki)	[[link picker logic	@tiptap/suggestion
Collaboration	Y.js CRDT sync (Phase 2)	@tiptap/extension-collaboration
CollaborationCursor	Per-user cursor + colour (Phase 2)	@tiptap/extension-collaboration-cursor
LivePreviewDecorations	Cursor-aware syntax reveal — custom	Custom ProseMirror Plugin
WikiLink	[[...]] node type — renders as link, editable	Custom Tiptap Node
Callout	Info/warn/danger block node	Custom Tiptap Node
FileAttachment	File block node	Custom Tiptap Node

10. Delivery Milestones

These milestones assume one frontend engineer. Timeline is indicative.

Milestone	Deliverable	Target
M1	Editor shell + Tiptap setup + StarterKit + basic markdown shortcuts + sidebar	Week 1
M2	LivePreviewDecorations plugin — all inline marks cursor-aware	Week 2
M3	Bubble menu + slash command menu + all block types	Week 3
M4	Wiki links (node + picker), math (KaTeX), table editing	Week 4
M5	Deep Tutor panel + slash AI commands + image/file blocks	Week 5
M6	Polish, edge cases, Markdown import/export, performance	Week 6
M7 (Phase 2)	Hocuspocus collab backend + Y.js sync + cursor presence	Week 8–10
M8 (Phase 2)	Graph view + backlinks panel	Week 11–12

11. Open Questions

- **Storage format** — Store notes as markdown (.md) or Tiptap JSON? Markdown is portable but loses Tiptap-specific node metadata. Recommended: Tiptap JSON as source of truth, .md as export format.

  Ans: Store notes as markdown-esque files with custom notations for the custom blocks and notes.
- **Math input UX** — Should math blocks be edited as raw LaTeX or via a visual equation editor?

  Ans: Visual editor
- **Offline support** — Y.js supports offline with sync-on-reconnect.

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